
Lecture 10: Emission Mechanisms of Continuous Radiation

Chin-Fei Lee

`cflee@asiaa.sinica.edu.tw`

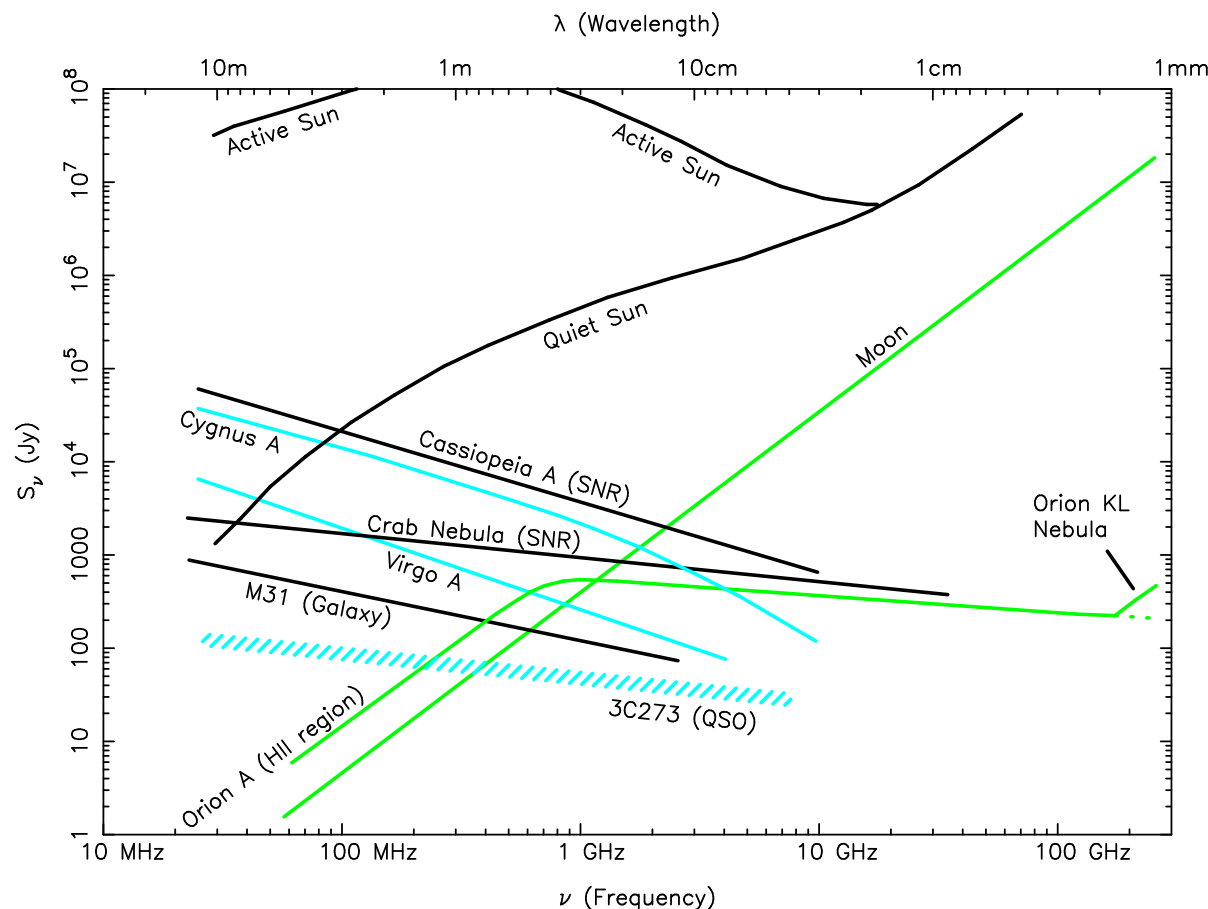
Nature of Radio Sources



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It may appear similar to the starry sky, but in fact most of the point-like objects are not stars, but luminous radio galaxies billions of light years away. The larger sources are ionized clouds of hydrogen, or supernova remnants.

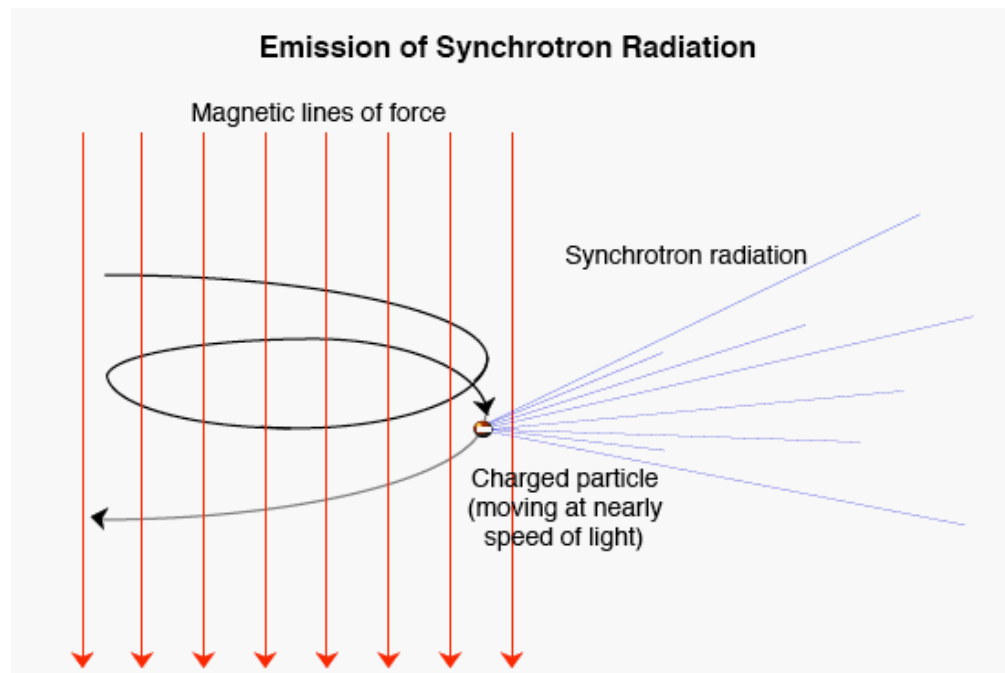
Nature of Radio Sources



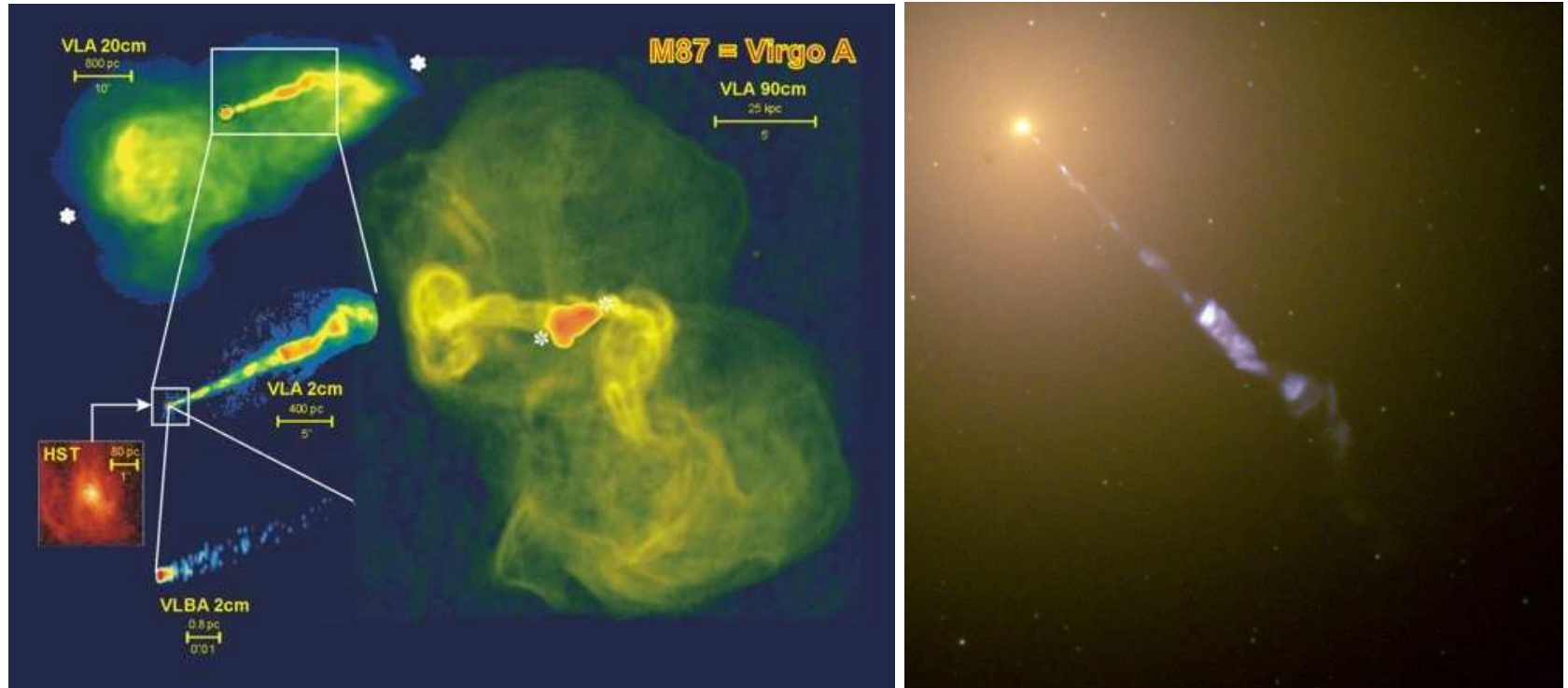
The spectral energy distributions of various radio sources. The Moon, the quiet Sun and (at lower ν) the HII region Orion A (optically thin above 1 GHz) are examples of **Black Bodies**. Close to 300 GHz there is **additional emission from dust** in the molecular cloud Orion KL Nebula. The active Sun, SNRs such as Cassiopeia A, the radio galaxies Cygnus A, Virgo A and the Quasi Stellar radio source (Quasar or QSO) 3C273 are **nonthermal emission**. The hatching indicates rapid time variability. Quasars and radio galaxies are likely the same kind of objects, namely both with AGNs at the centers.

Synchrotron Radiation

In 1950, Alfven and Herlofson proposed that the nonthermal radiation is emitted by [relativistic electrons moving in intricately "tangled" magnetic fields](#) of extended coronas believed to surround certain kinds of stars. The radiation was given the name “synchrotron radiation” since it was first discovered to be emitted by [electrons in a synchrotron accelerator](#). Synchrotron radiation is similar to [cyclotron radiation](#), but from ultrarelativistic electrons.



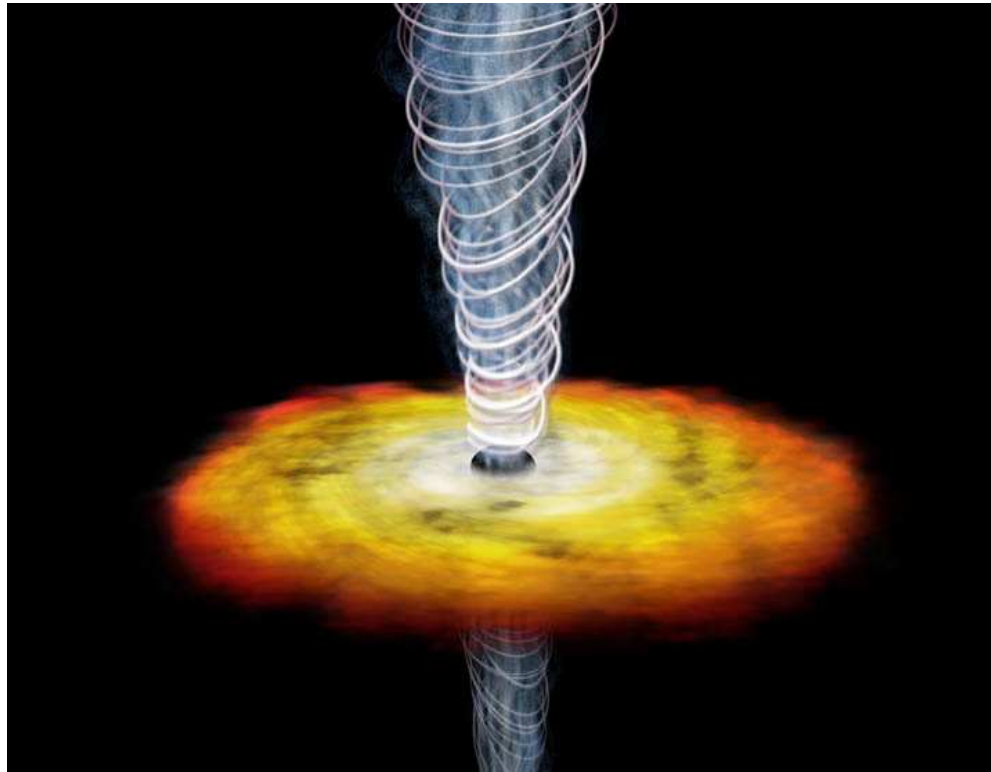
Radio Galaxy: M87 Virgo A



(Right) HST image of a 5000 lt-yr long jet being ejected from the [active galactic nucleus \(AGN\)](#) of the radio galaxy M87 (Virgo A). The synchrotron radiation of the jet (blue) contrasts with the starlight from the host galaxy (yellow). It is emitted by high-energy electrons spiraling along magnetic field lines. It was first detected in 1956 by Geoffrey R. Burbidge in M87, confirming a prediction by Hannes Alfvén and Nicolai Herlofson in 1950, and Iosif S. Shklovskii in 1953.

AGNs have two classes: [radio-quiet](#) and [radio-loud](#). In the radio-loud objects a contribution from the jet(s) and the lobes they inflate dominates the luminosity of the AGN, at least at radio wavelengths. Radio-quiet objects are simpler since jet and jet-related emission can be neglected.

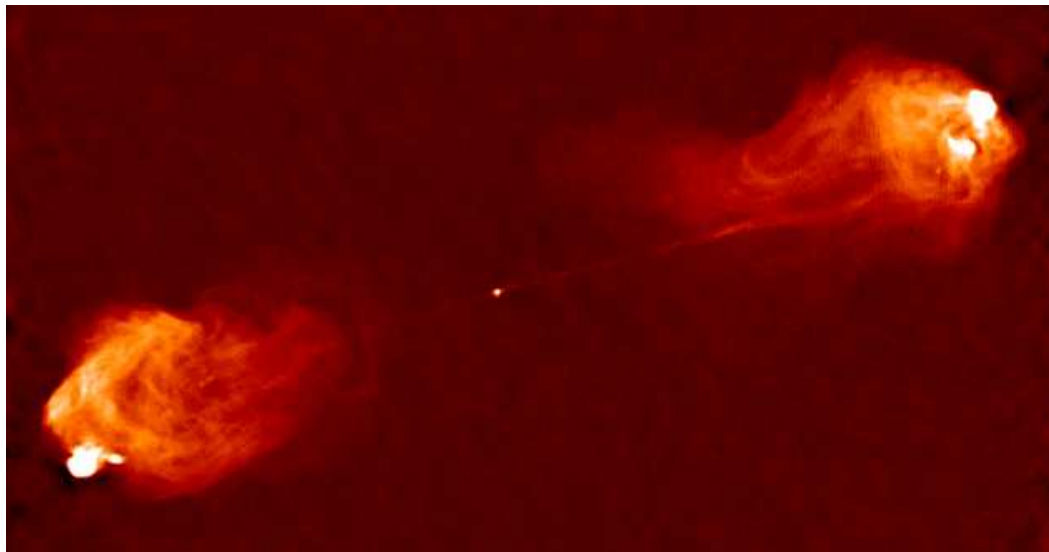
Formation of AGN jets



Differential rotation twists up field into toroidal component. Greatest field enhancement is at disk inner edge. Magnetic pressure gradient (dB_{ϕ}^2/dZ) accelerates plasma out of system. Magnetic tension [hoop stress] (B_{ϕ}^2/R) pinches and collimates the outflow into a jet. Outflow jet speed is of order the escape velocity from the inner edge of the torus ($V_{\text{jet}} \sim V_{\text{Alfven}} \sim V_{\text{esc}}$). Jet direction is along the rotation axis.

Radio Galaxy Cygnus A

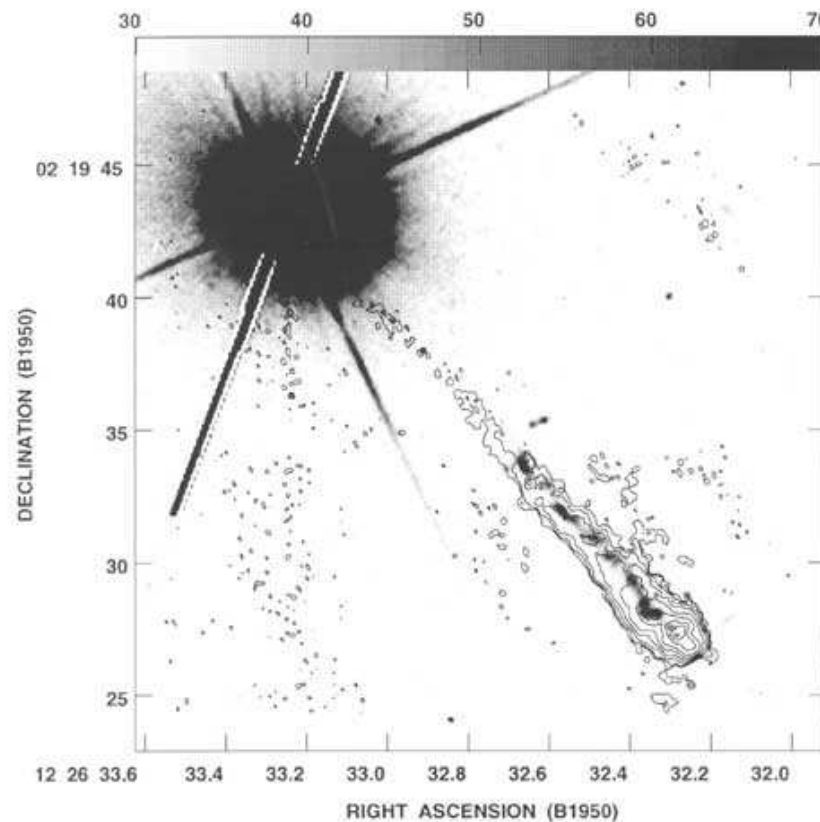
Cygnus A: Strongest extragalactic radio source in the sky. The 1954 identification of this source with an extremely distant (redshift $z \approx 0.057$, corresponding to a distance $D \sim 240$ Mpc) galaxy stunned radio astronomers, who immediately recognized that such a luminous radio source ($\approx 10^{45} \text{ erg s}^{-1} = 10^{38} \text{ W}$) could be detected almost anywhere in the universe. The angular extent of this source, about 100 arcsec, implies a linear extent of about 100 kpc, much larger than the host galaxy. The energy source is clearly not stars. Gravitational energy released by matter accreting onto a supermassive ($M \sim 10^9 M_{\odot}$) black hole in the center of the host galaxy powers this and other luminous extragalactic radio sources.



VLA image: The bright central core coincides with a supermassive black hole that accelerates the relativistic electrons along two jets that terminate in lobes well outside the host galaxy.

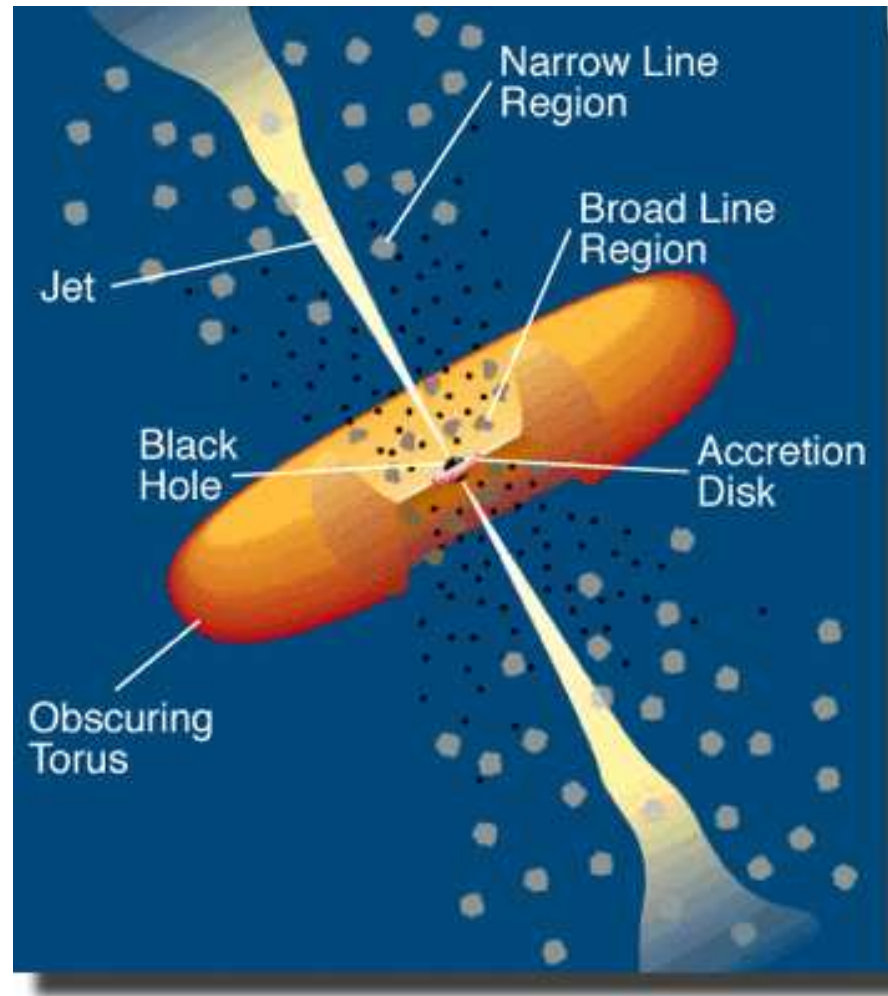
Quasar 3C 273

The bright radio source 3C 273 was identified with the first quasar at an even higher redshift, $z=0.16$. Such quasars appear to be radio galaxies in an especially active state, when **visible light from the region of the black hole overwhelms the starlight from the host galaxy and makes the quasar look like a bright star.**



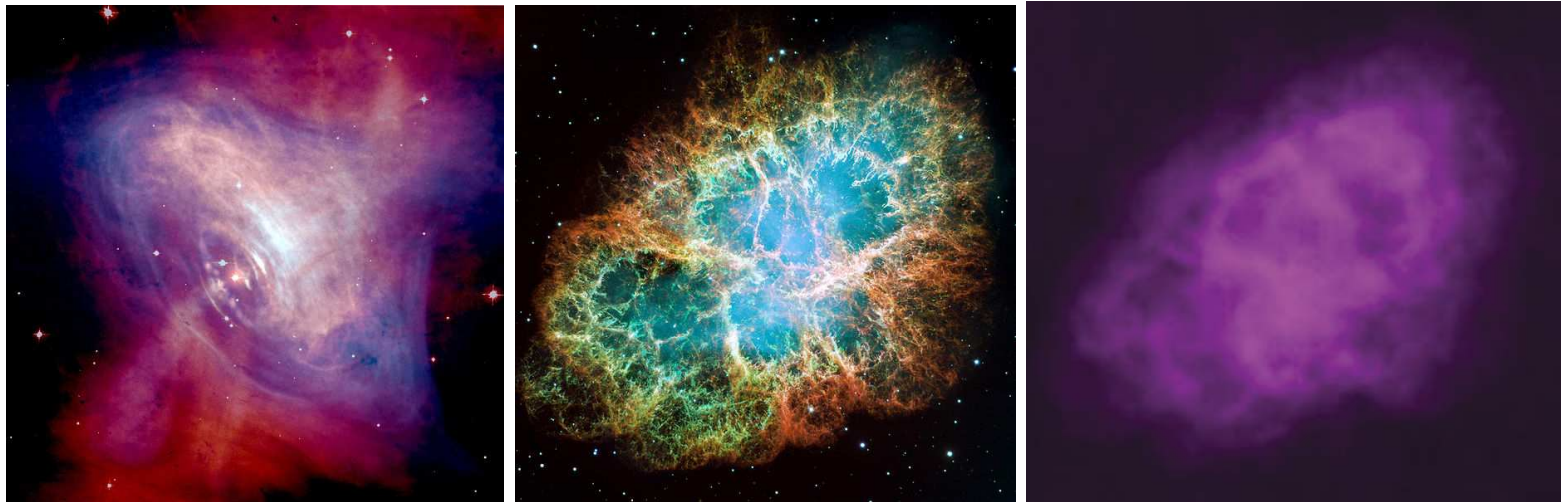
This HST gray-scale image of the quasar 3C273 shows radio contours superimposed on the optical jet emission.

AGN model



Supernova Explosion Crab Nebula

From the Sung-shih [Annals, of the Sung Dynasty]: "On the 1st year of the Chi-ho reign period, 5th month, chi-chou (day) [July 4, 1054], a guest star appeared approximately several inches to the south-east of Tian-kuan [Taurus]. After a year and more it gradually vanished."

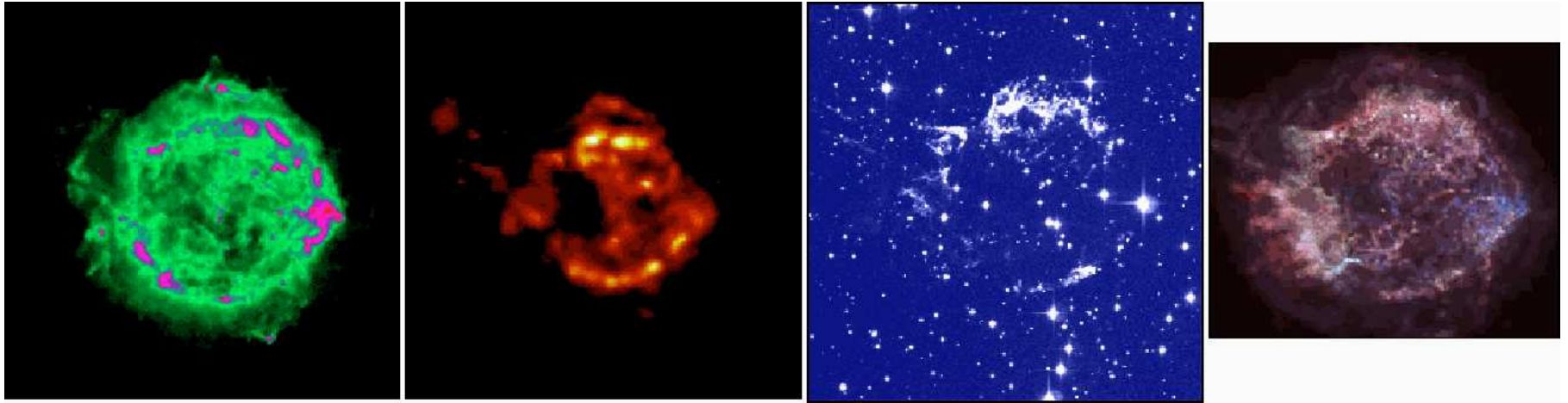


(Left) Optical data from Hubble (red) and X-ray images from Chandra X-ray Observatory (blue). At the center of the nebula lies the Crab Pulsar, a rapidly rotating neutron star, which emits pulses of radiation from gamma rays to radio waves with a spin rate of 30.2 times per second. Its powerful B field concentrates its radiation emission into a narrow beam. The nebula was the first astronomical object identified with a historical supernova explosion. (Mid) By HST WFPC2 (UV-Infrared). The bluish glow from the central region of the nebula is probably due to synchrotron radiation from the pulsar wind of which the electrons are trapped in the strong B field around the pulsar. (Right) Radio image by VLA showing the synchrotron radiation in radio.

Supernova Explosion Cassiopeia A

A star that died in ~ 1700 .

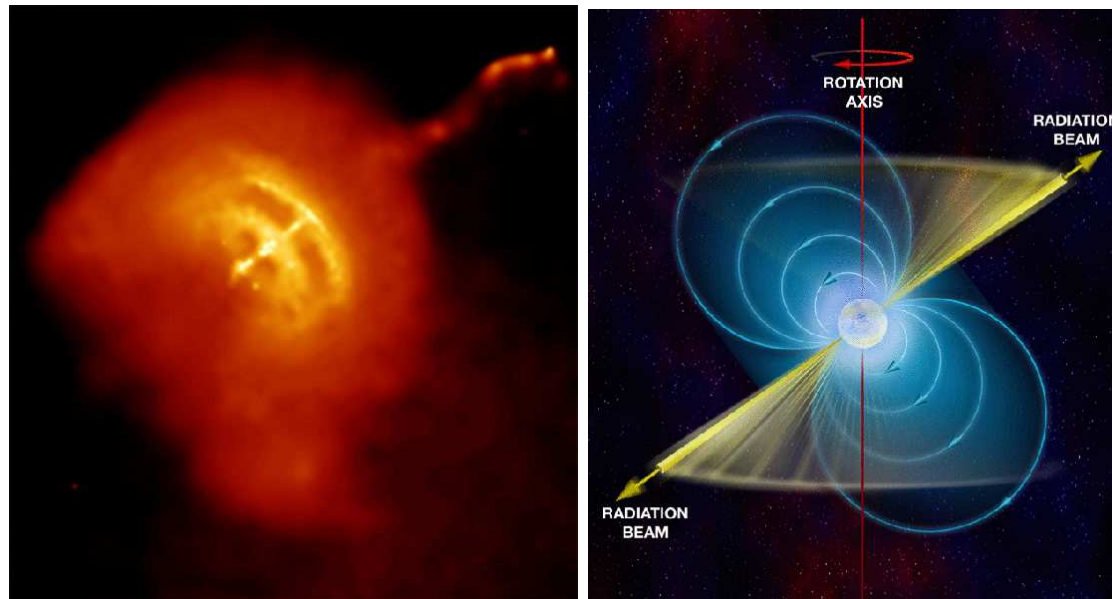
From left to right: Radio, Infrared, Optical, to X-ray.



Pulsars: Pulsating Stars

Pulsars are highly **magnetized, rotating neutron stars** that emit a beam of electromagnetic radiation. The observed periods of their pulses range from **1.4 milliseconds to 8.5 seconds**. The radiation can only be observed when the beam of emission is pointing towards the Earth. This is called the **lighthouse effect** and gives rise to the pulsed nature that gives pulsars their name.

The first pulsar was observed in July 1967 by Jocelyn Bell Burnell and Anthony Hewish. Initially baffled as to the seemingly unnatural regularity of its emissions, they dubbed their discovery LGM-1, for "little green men" (a name for intelligent beings of extraterrestrial origin). The hypothesis that pulsars were beacons from extraterrestrial civilizations was never serious, but some discussed the far-reaching implications if it turned out to be true.



(Left) The Vela Pulsar and its surrounding pulsar wind nebula. (Right) The lighthouse model.

Supernova explosion

Stellar Evolution of high-mass stars:



Synchrotron Radiation

Since the electrons that emit the synchrotron radiation are highly relativistic, [Lorentz transformations \(special theory of relativity\)](#) must be used to describe their motion.

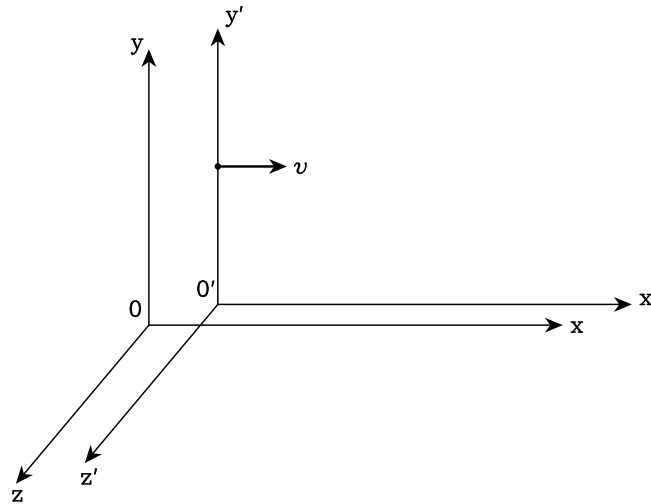
The Lorentz transformation converts between two different observers' measurements of space and time, where one observer is in constant motion with respect to the other. With this, we can derive our solution in one frame (i.e., an inertial frame moving with the electron and thus nonrelativistic physics can be adopted) and convert it to another frame (observer's frame).

There are two postulates in special theory of relativity:

1. The laws of nature are the same in any two frames of reference that are in uniform relative motion
2. The speed of light, c , is constant in all such frames.

Lorentz Transformation

Two inertial frames K and K' with a relative velocity v along the x-axis:



The origins are assumed to coincide at $t = 0$. If a pulse of light is emitted at the origin at $t = 0$, each observer will see an expanding sphere centered on his own origin because the speed of light is c and constant:

$$x^2 + y^2 + z^2 - c^2 t^2 = 0 \qquad x'^2 + y'^2 + z'^2 - c^2 t'^2 = 0$$

In another word, the evolutions of space and time are bonded by the speed of light $\Rightarrow$ **4-dimensional spacetime** and lengths and times must be transformed together from one frame to another. If space is homogeneous, then the Lorentz transformation must be a linear transformation.

Lorentz Transformation

The transformation can be thought of some kind of rotation in spacetime with

$$\begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix}$$

Or

$$\begin{aligned} t &= \gamma(t' + \beta \frac{x'}{c}) \\ x &= \gamma(x' + vt') \\ y &= y' \\ z &= z' \end{aligned}$$

where

$$\beta = \frac{v}{c} \quad \text{and} \quad \gamma^2 - \beta^2\gamma^2 = 1 \Rightarrow \gamma = (1 - \beta^2)^{-1/2}$$

As a result, the lengths and times are distorted when seen from other observer. In classical physics (Galilean relativity), the only conversion is $x = x' + vt'$ with $t = t'$.

Lorentz Transformation

In addition, we have

$$\frac{dt}{dt'} = \gamma(1 + \beta \frac{u'_x}{c}) \equiv \gamma\sigma \quad \text{with} \quad \sigma = 1 + \beta \frac{u'_x}{c}$$

with $u'_x = dx'/dt'$. Note that this gives the [time dilation](#), e.g., a clock at rest (i.e., $u'_x = 0$) in the system K' is seen in the system K with $\Delta t = \gamma\Delta t'$. Then the velocity conversion is:

$$\begin{aligned} u_x &= \frac{dx}{dt} = \frac{dx}{dt'} \frac{dt'}{dt} = \gamma(u'_x + v) \frac{dt'}{dt} = \frac{1}{\sigma}(u'_x + v) \\ u_y &= \frac{dy}{dt} = \frac{dy}{dt'} \frac{dt'}{dt} = u'_y \frac{dt'}{dt} = \frac{1}{\gamma\sigma} u'_y \\ u_z &= \frac{dz}{dt} = \frac{dz}{dt'} \frac{dt'}{dt} = u'_z \frac{dt'}{dt} = \frac{1}{\gamma\sigma} u'_z \end{aligned}$$

with $u'_y = dy'/dt'$, $u'_z = dz'/dt'$. The transformation can be formulated in terms of the velocity components $\parallel$ and $\perp \mathbf{v}$:

$$u_{\parallel} = \frac{u'_{\parallel} + v}{1 + \beta \frac{u'_{\parallel}}{c}} \quad \text{and} \quad u_{\perp} = \frac{u'_{\perp}}{\gamma(1 + \beta \frac{u'_{\parallel}}{c})}$$

When $v \rightarrow c$, we have $\beta \rightarrow 1$ and $\gamma \gg 1$, then $u_{\perp} \ll u'_{\perp} \Rightarrow$ [beaming effect seen in frame K](#).

Moreover, for radiation emitted in K' , we have $u'_{\parallel} = u'_{\perp} = c$, then $u_{\parallel} = c = u'_{\parallel}$ and $u_{\perp} = \frac{c}{2\gamma}$

Lorentz Transformation

As for the acceleration, we have

$$\begin{aligned}a_{\parallel} &= \frac{du_{\parallel}}{dt} = \frac{du_{\parallel}}{dt'} \frac{dt'}{dt} = \gamma^{-3} \sigma^{-3} a'_{\parallel} \\a_{\perp} &= \frac{du_{\perp}}{dt} = \frac{du_{\perp}}{dt'} \frac{dt'}{dt} = \gamma^{-2} \sigma^{-3} \left(\sigma a'_{\perp} - \beta \frac{u'_{\perp}}{c} a'_{\parallel} \right)\end{aligned}$$

For the special case when the particle is at rest in system K' , that is, that initially $u'_{\parallel} = u'_{\perp} = 0$, we have $\sigma = 1$ and

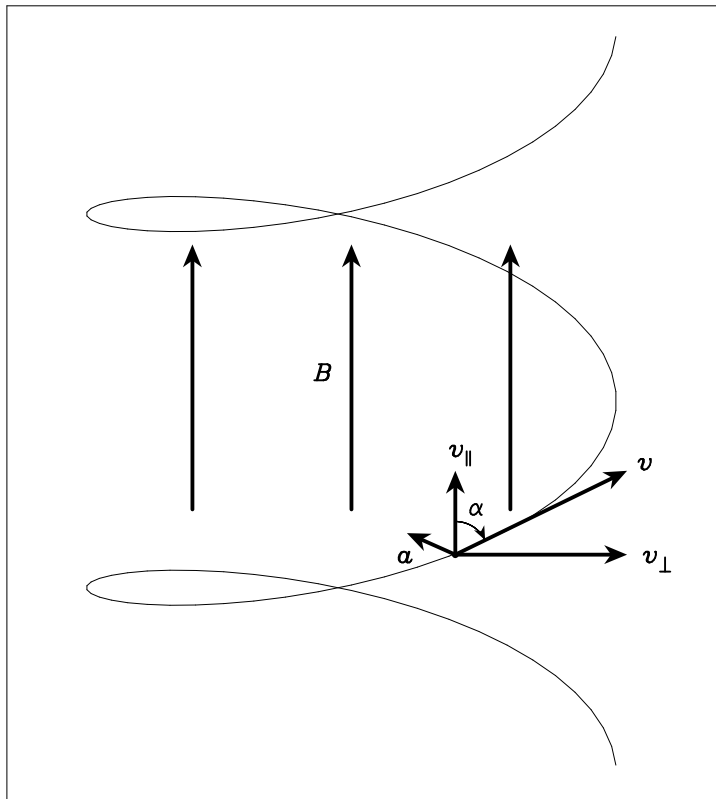
$$a_{\parallel} = \gamma^{-3} a'_{\parallel} \quad \text{and} \quad a_{\perp} = \gamma^{-2} a'_{\perp}$$

Synchrotron Radiation of 1 Electron

The motion of an electron in a (homogeneous) magnetic field is governed by ($m \Rightarrow \gamma m$)

$$\frac{d}{dt}(\gamma m \mathbf{v}) = \frac{e}{c}(\mathbf{v} \times \mathbf{B})$$

where m is the rest mass and γm is equivalent mass.



Projecting $\mathbf{v}$ into two components,
 $\mathbf{v}_{\parallel}$ parallel to $\mathbf{B}$ and $\mathbf{v}_{\perp}$ perpendicular to $\mathbf{B}$, we have

$$\frac{d\mathbf{v}_{\parallel}}{dt} = 0 \quad \text{and} \quad \frac{d\mathbf{v}_{\perp}}{dt} = \frac{e}{\gamma m c}(\mathbf{v}_{\perp} \times \mathbf{B})$$

So, $\mathbf{v}_{\parallel} = \text{constant}$.

Since the force is perpendicular to $\mathbf{v}_{\perp}$, the motion is a uniform circular motion with a constant orbital velocity $v_{\perp} \Leftrightarrow$ **relativistic cyclotron (gyration) radiation**. The path followed by the electron is a helix winding around $\mathbf{B}$ with the constant **pitch angle** (the angle betw. the field and velocity) $\tan \alpha = \frac{|\mathbf{v}_{\perp}|}{|\mathbf{v}_{\parallel}|}$.

Synchrotron Radiation of 1 Electron

The frequency of gyration:

$$\omega_B = \frac{eB}{\gamma mc} \equiv \frac{\omega_G}{\gamma} \quad \text{with} \quad \omega_G = \frac{eB}{mc}$$

Since the equivalent "mass" is a factor of γ the rest mass, the frequency is smaller by a factor of γ . In the K frame, the acceleration is $a_{\perp} = \omega_B v_{\perp}$.

Now if we select an inertial frame K' that moves with respect to rest frame K such that the electron is at rest at a certain time, then this particle will not remain at rest for long since its acceleration is not zero. In frame K' , the electron is nonrelativistic and radiates according to the Larmor formula:

$$P' = \frac{2e^2}{3c^3} a_{\perp}'^2$$

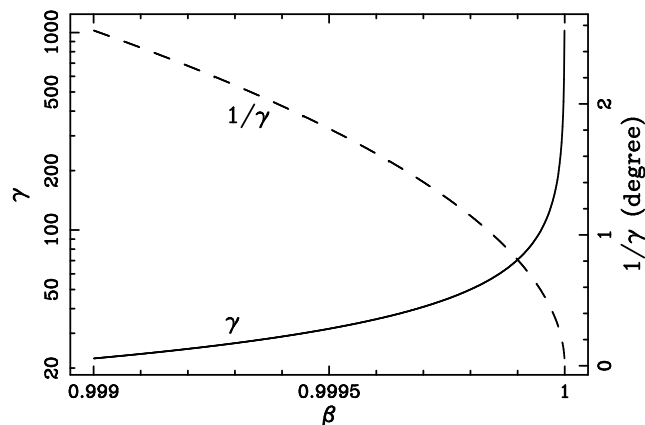
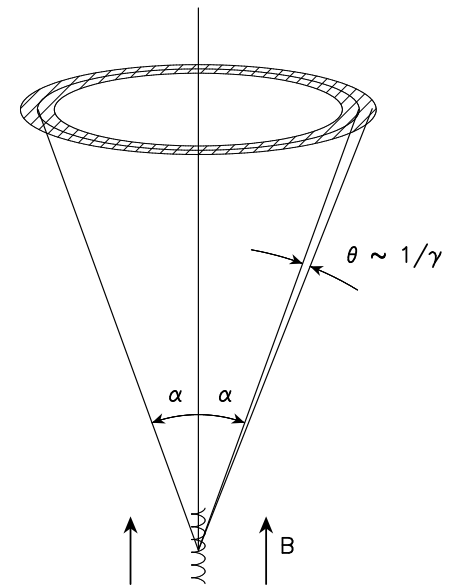
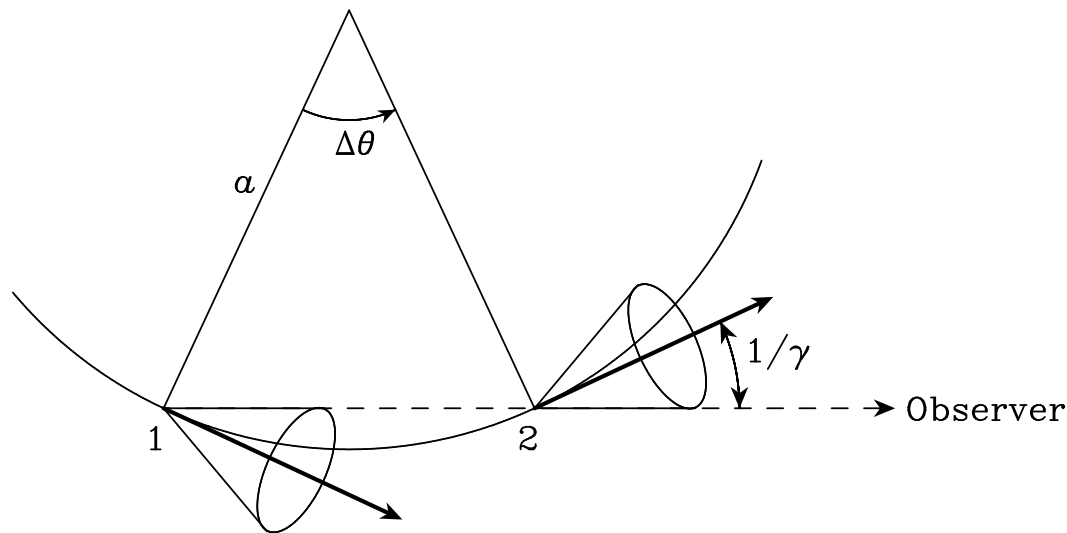
The total power radiated in frame K (observer's frame) is then (with $a_{\perp}' \Rightarrow \gamma^2 a_{\perp}$)

$$P = P' = \frac{2e^2}{3c^3} \gamma^4 a_{\perp}^2 = \frac{2e^2}{3c^3} \gamma^4 \frac{e^2 B^2}{\gamma^2 m^2 c^2} v_{\perp}^2 = 2\sigma_T c \beta_{\perp}^2 \gamma^2 U_B$$

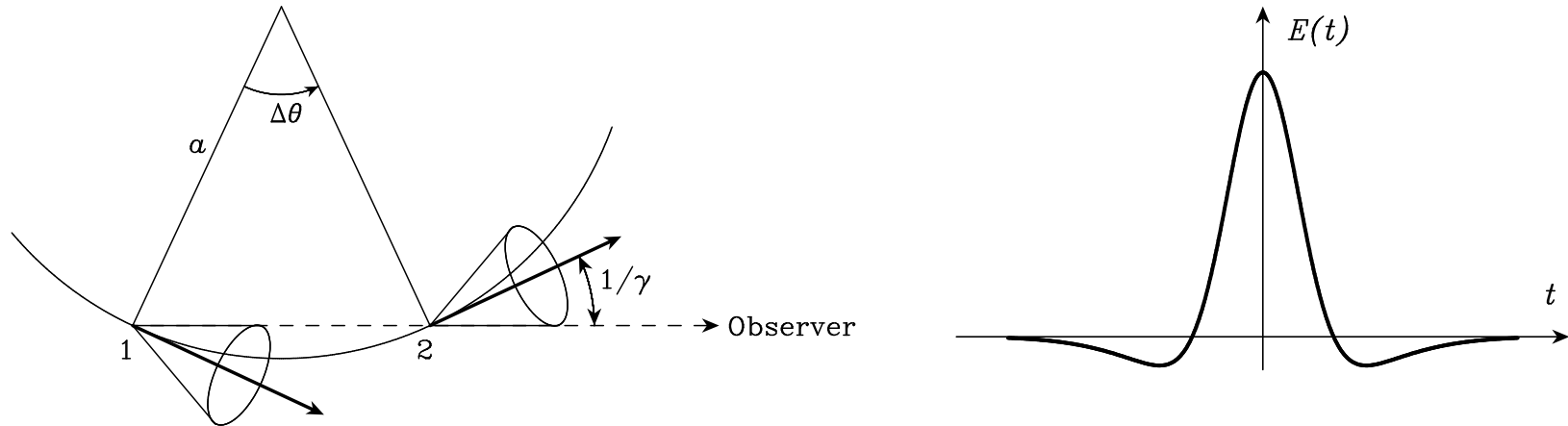
where $\sigma_T = \frac{8\pi r_0^2}{3} = 6.65 \times 10^{-25} \text{ cm}^2$ is the Thomson scattering cross section of electron with $r_0 = \frac{e^2}{mc^2}$ (classical electron radius), $\beta_{\perp} = \frac{v_{\perp}}{c}$, and magnetic energy density $U_B = \frac{B^2}{8\pi}$.

Beaming effect

In frame K' , the radiation pattern is similar to that of the Hertz dipole with the dipole (or *current I*) direction given by the acceleration direction of the electron. But in frame K (observer's frame) the radiation is shifted to the forward direction and confined to a cone with an angle of $1/\gamma$.



Beaming effect



The observer will see the pulse of radiation from points 1 and 2 along the electron's path.

$$\Delta t = t_2 - t_1 \approx \frac{\Delta \theta}{\omega_B \sin \alpha} = \frac{2/\gamma}{\omega_B \sin \alpha}$$

Let t_1^A and t_2^A be the **arrival times** of the radiation at the observer. The difference $t_2^A - t_1^A$ is less than $t_2 - t_1$ by an amount $\Delta s/c$ (**Doppler effect**) where $\Delta s = v(t_2 - t_1)$. Thus,

$$\Delta t^A = t_2^A - t_1^A = \Delta t \left(1 - \frac{v}{c}\right) = \Delta t \frac{\left(1 - \frac{v^2}{c^2}\right)}{1 + \frac{v}{c}} \approx \frac{\Delta t}{2\gamma^2} = \frac{1}{\gamma^3 \omega_B \sin \alpha}$$

Synchrotron Radiation

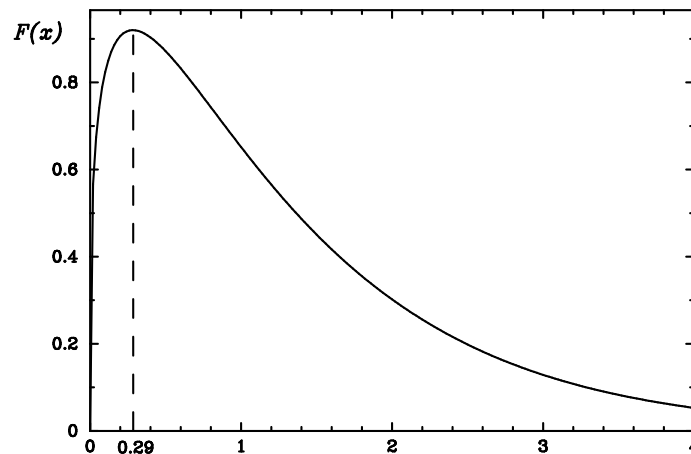
The spectrum of the radiation can be obtained by Fourier transform of the radiation pulse received. The emissivity is:

$$P(\nu) = \frac{\sqrt{3}e^3 B \sin \alpha}{mc^2} F\left(\frac{\nu}{\nu_c}\right)$$

where the critical frequency ν_c is defined by

$$\nu_c = \frac{3}{2}\gamma^3 \nu_B \sin \alpha = \frac{3}{2}\gamma^2 \nu_G \sin \alpha \quad \text{with} \quad \nu_G = \frac{\omega_G}{2\pi} = \frac{eB}{2\pi mc}$$

and $F(x) = x \int_x^\infty K_{5/3} dx$ with $K_{5/3}$ being the modified Bessel function of 5/3 order.



For an ensemble of electrons

From empirical evidence, cosmic ray data show that number density of electrons with energies between E and $E + dE$ (or γ and $\gamma + d\gamma$ since $E = \gamma mc^2$) can be approximately expressed in the form

$$N(E)dE = CE^{-p}dE \quad E_1 < E < E_2$$

or

$$N(\gamma)d\gamma = C\gamma^{-p}d\gamma \quad \gamma_1 < \gamma < \gamma_2$$

The quantity C can vary with pitch angle and the like. The emissivity (power per unit frequency interval per unit volume and per unit solid angle) of the relativistic electrons is given by

$$j_\nu = C \int_{\gamma_1}^{\gamma_2} P(\nu)\gamma^{-p}d\gamma \propto \int_{\gamma_1}^{\gamma_2} F\left(\frac{\nu}{\nu_c}\right)\gamma^{-p}d\gamma$$

Let $x \equiv \nu/\nu_c$, noting $\nu_c \propto \gamma^2$, then

$$j_\nu \propto \nu^{-(p-1)/2} \int_{x_1}^{x_2} F(x)x^{(p-3)/2}dx \propto \nu^{-(p-1)/2}$$

Putting in all the quantities we have

$$j_\nu = \frac{\sqrt{3}e^3CB \sin \alpha}{mc^2(p+1)} \Gamma\left(\frac{3p+19}{12}\right) \Gamma\left(\frac{3p-1}{12}\right) \left(\frac{2\pi mc\nu}{3eB \sin \alpha}\right)^{-(p-1)/2}$$

Absorption

Synchrotron emission is accompanied by [absorption](#), in which a photon interacts with a charge in a magnetic field and is absorbed, giving up its energy to the charge. Another process that can occur is [stimulated emission or negative absorption](#), in which a particle is induced to emit more strongly into a direction and at a frequency where photons already present. This absorption can become important in compact, high-intensity radio sources at low frequencies when the optical depth becomes large.

The derivation of the absorption coefficient is complicated. Here is the result:

$$\alpha_\nu = \frac{\sqrt{3}e^3}{8\pi m} \left(\frac{3e}{2\pi m^3 c^5} \right)^{p/2} C(B \sin \alpha)^{(p+2)/2} \Gamma\left(\frac{3p+2}{12}\right) \Gamma\left(\frac{3p+22}{12}\right) \nu^{-(p+4)/2}$$

The source function can be found from

$$S_\nu = \frac{j_\nu}{\alpha_\nu} \propto \nu^{5/2}$$

The source function is a power with an index $-\frac{5}{2}$, independent of the value of p . This index is not equal to -2 , the Rayleigh-Jeans value, because the emission is nonthermal.

Spectrum of Synchrotron Radiation

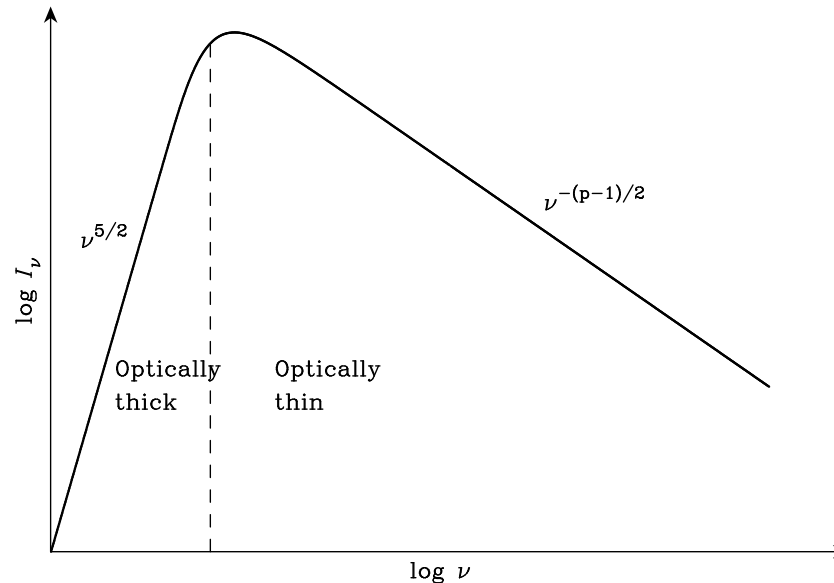
The radiative transfer can be written as

$$I_\nu(\tau_\nu) = S_\nu(1 - e^{-\tau_\nu})$$

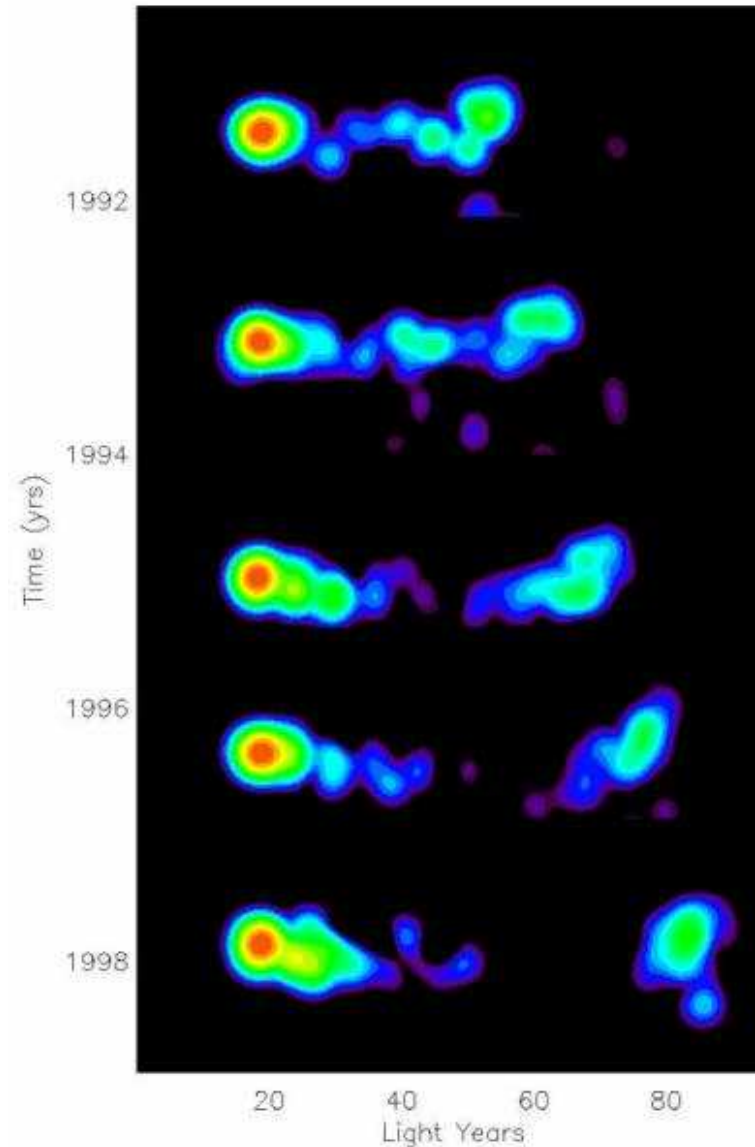
and thus

$$I_\nu(\tau_\nu) = \begin{cases} S_\nu \propto \nu^{5/2} & \text{for } \tau_\nu \gg 1 \\ S_\nu \tau_\nu \approx j_\nu L \propto \nu^{-(p-1)/2} & \text{for } \tau_\nu \ll 1 \end{cases}$$

with L being the depth of the emission region along the line of sight.



Superluminal Motion: 3C 273



E.g., Radio jet of 3c 273 moves for 25 light-years in 7 yrs!!

Superluminal Motion + Beaming

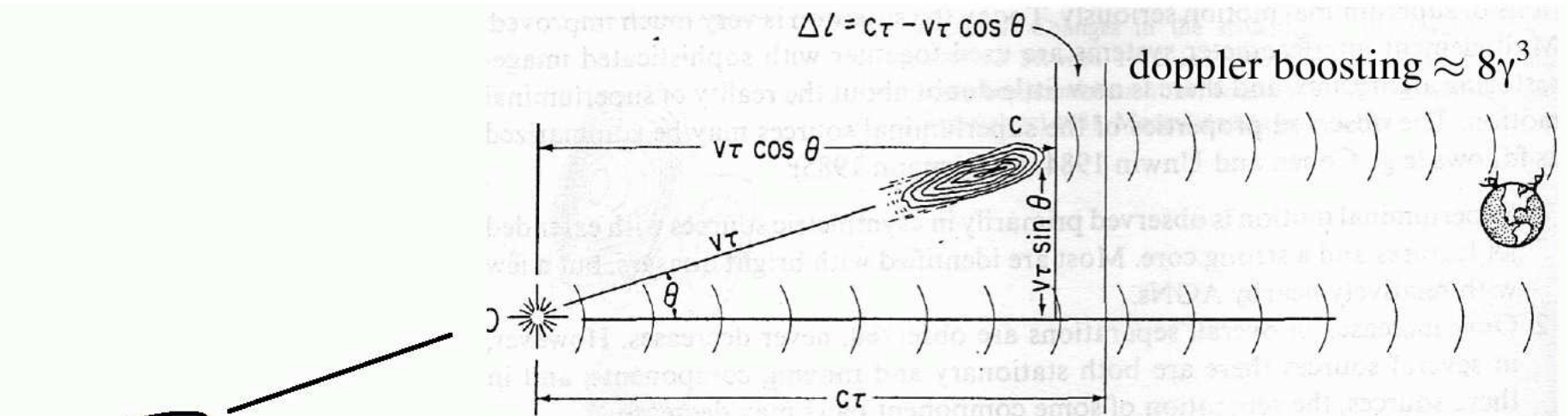


Fig. 13.8. Apparent superluminal motion results when the radiating source is moving so fast that it nearly catches up with its own radiation. Assume that a radiating plasma cloud is ejected from the origin, O , with a velocity v in a direction θ with respect to the line of sight. After a time t , the cloud has moved a distance vt . The motion, projected along the line of sight is $vt \cos \theta$, and projected perpendicular to the line of sight, $vt \sin \theta$. A distant observer sees the emission delayed by a time $t(c - v \cos \theta) = ct(1 - \beta \cos \theta)$ compared to the “signal” radiated when the cloud was at O . The apparent transverse velocity seen by the observer is then $(vt \sin \theta) / [ct(1 - \beta \cos \theta)] = \beta \sin \theta / (1 - \beta \cos \theta)$.

$$v_{app} = \frac{v \sin \theta}{1 - \beta \cos \theta}$$

It has a max. value of $\gamma\beta c = \gamma v \sim \gamma c$ when $\cos \theta = \beta$ at which $dv_{app}/d\theta = 0$. E.g., if $\beta = 0.99$ then $\gamma \sim 7$, and the max. apparent velocity is $\sim 7c$ when $\theta \sim 8^\circ$.

Doppler Boosting

(From Kellermann, Kovalev, Lister, et. al 2007). Due to relativistic effects, we also observe apparent luminosities and brightness temperatures which are related to the corresponding intrinsic quantities in the AGN rest frame through the **Doppler factor**, δ .

The apparent luminosity and the apparent brightness temperature are

$$L_{\text{app}} = \delta^n L_{\text{int}}, \quad T_{\text{app}} = \delta T_{\text{int}}$$

where n depends on the geometry and spectral index and is typically in the range between 2 and 3. The Doppler factor is

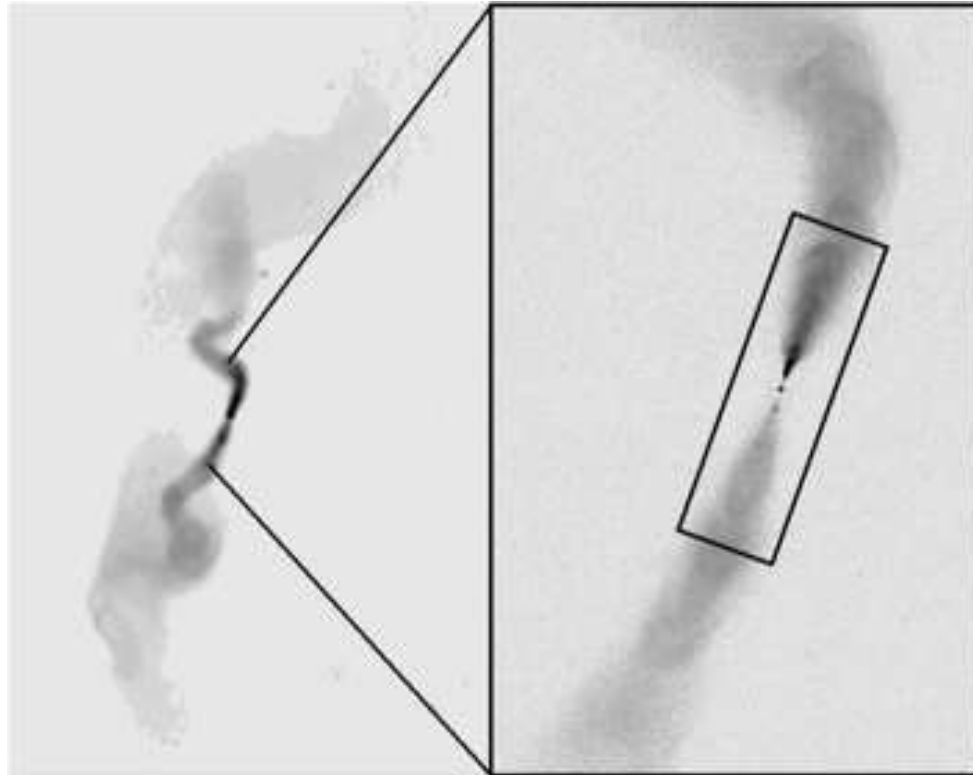
$$\delta = \frac{1}{\gamma(1 - \beta \cos \theta)}$$

For e.g., $v \approx c$ then we have $1 - \beta \approx 1/(2\gamma^2)$ and if $\theta \approx 0$ then

$$\delta \approx \frac{2\gamma^2}{\gamma} = 2\gamma \quad \Rightarrow \quad L_{\text{app}} \approx 8\gamma^3 L_{\text{int}} \quad (\text{if } n = 3), \quad T_{\text{app}} \approx 2\gamma T_{\text{int}}$$

$\Rightarrow$ Doppler Boosting: An effect, predicted by the special theory of relativity, that enhances the radiation from material that is moving toward the observer at nearly the speed of light, and hides material moving in the opposite direction at such speeds.

Doppler Boosting



Binary Pulsars

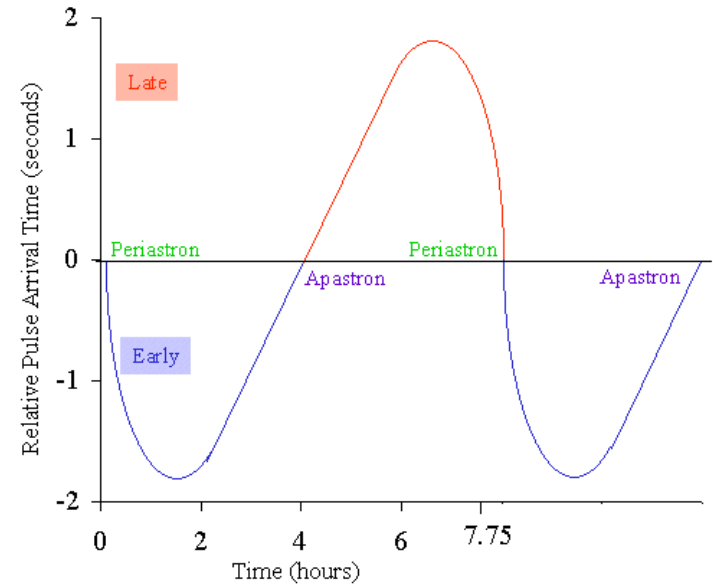
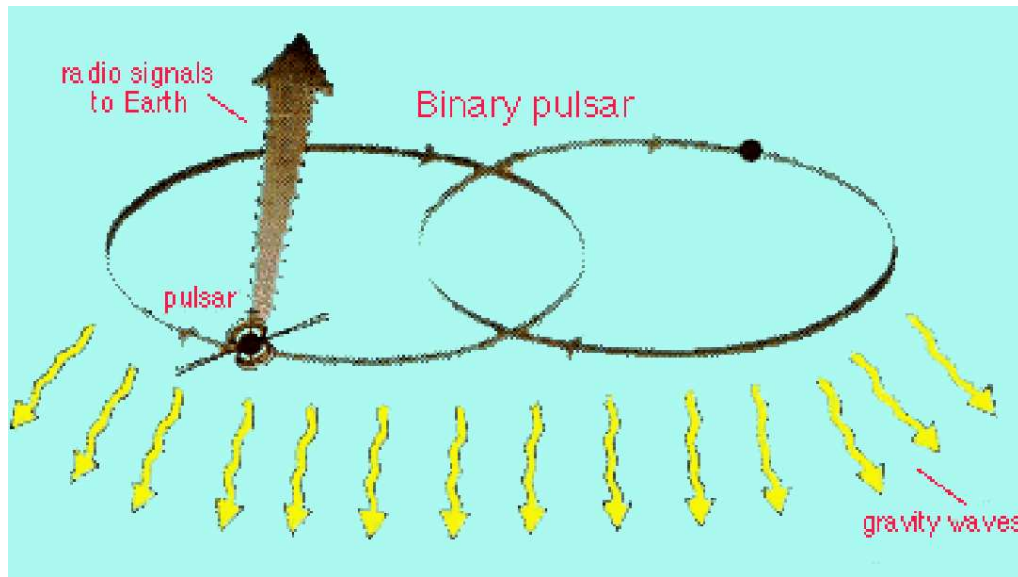
In 1993, the Nobel Prize in Physics was awarded to Russell Hulse and Joseph Taylor of Princeton University for their 1974 discovery of a pulsar, designated PSR1913+16, in a binary system, in orbit with another star around a common center of mass.

Using the Arecibo 305m antenna, Hulse and Taylor detected pulsed radio emission and thus identified the source as a pulsar, a rapidly rotating, highly magnetized neutron star. The neutron star rotates on its axis 17 times per second; thus the pulse period is 59 milliseconds.

After timing the radio pulses for some time, Hulse and Taylor noticed that there was a systematic variation in the arrival time of the pulses. Sometimes, the pulses were received a little sooner than expected; sometimes, later than expected. These variations changed in a smooth and repetitive manner, with a period of 7.75 hours. They realized that such behavior is predicted if the pulsar were in a binary orbit with another star.

The stars are believed to be nearly equal in mass, about 1.4 solar masses. As shown in the figure here, the orbits are quite eccentric. The minimum separation at periastron is about 1.1 solar radii; the maximum separation at apastron is 4.8 solar radii.

Binary Pulsar: PSR 1913+16



(Right Figure from Weisberg et al. 1981).

The pulsar arrival times vary as the pulsar moves through its orbit. When the pulsar is on the side of its orbit closest to the Earth, the pulses arrive more than 3 seconds earlier than they do when it is on the side furthest from the Earth. The difference is caused by the shorter distance from Earth to the pulsar when it is on the close side of its orbit. The difference of 3 light seconds implies that the orbit is about 1 million kilometers across.

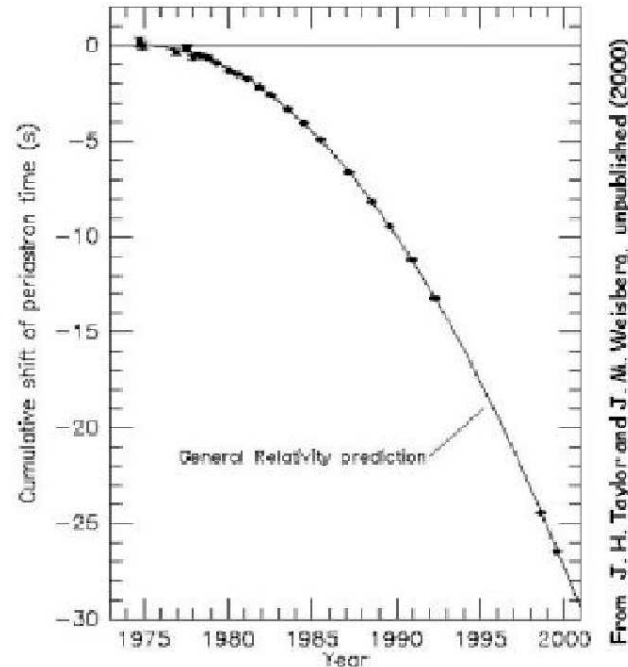
Gravitaional Wave

Gravitational wave is a fluctuation in the curvature of spacetime which propagates as a wave, traveling outward from a moving object or system of objects. Gravitational radiation is the energy transported by these waves. Important examples of systems which emit gravitational waves are binary star systems, where the two stars in the binary system are either white dwarfs, neutron stars, or black holes. Although gravitational radiation has not yet been directly detected, it has been indirectly shown to exist. This was the basis for the 1993 Nobel Prize in Physics, awarded for measurements of the [Hulse-Taylor binary system PSR 1913+16](#).

In Einstein's theory of general relativity, gravity is postulated to be a result of the curvature of spacetime. This curvature is caused by the presence of massive objects. As massive objects move around in spacetime, the curvature will change to reflect the changed locations of those objects. If the objects move around in a certain way, ripples in spacetime can spread outward "like ripples on the surface of a pond". These ripples are gravitational waves. [In general terms, gravitational waves are radiated by objects whose motion involves acceleration. Another way of looking at it is by analogy to electromagnetic waves: gravitational waves are the radiating or "shining" component of the gravitational field.](#)

Imagine a simple system of two masses such as the Earth-Sun system moving slowly compared to the speed of light. Assume that these two masses orbit each other in a circular orbit in the x-y plane. Such an orbit represents a changing quadrupole moment and will thus give off gravitational waves.

Gravitaional Wave: PSR 1913+16



The orbit has decayed since the binary system was initially discovered, in precise agreement with the loss of energy due to gravitational waves predicted by Einstein's General Theory of Relativity. The rate of decrease of orbital period is 0.0000765 seconds per year, the rate of decrease of semimajor axis is 3.5 meters per year.

$$\frac{dE}{dt} = -\frac{32}{5} \frac{G}{c^5} \left(\frac{M_1 M_2}{M_1 + M_2} \right)^2 R^4 \omega^6 \approx 2^7 \frac{M_1}{M_2} \left(\frac{v}{c} \right)^5 \frac{E_{\text{kin}}}{T}$$

where G is the gravitational constant and E_{kin} , T and v are the kinetic energy, the period time and velocity of the first mass.